When Does Detect-and-Reweight Work? Epoch Confounds and Adaptive Reweighting in Continued LM Pretraining

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Abstract

As synthetic text reaches an estimated 20–60% of web corpora, the detect-and-reweight paradigm (downweighting likely-synthetic samples via $w(x) = (1 - q(x))^b$) has gained traction, but its effectiveness conditions remain poorly understood. Through controlled experiments (Qwen3-0.6B continued pretraining, 132M FineWeb tokens, 3 seeds), we analyze when, why, and how detect-and-reweight works.

We identify three findings. First, an epoch-exposure confound in detect-and-remove evaluation: at matched content epochs, the advantage shrinks from 0.396 to 0.017 perplexity (N=3), revealing that >95% of published gains (relative to adaptive reweighting) reflect data repetition rather than filtering quality. Second, effective reweighting requires both sufficient discrimination and good calibration from the detector; a high-AUC but poorly calibrated detector (ECE = 0.131) degrades performance below uniform, while a lightweight calibrated alternative (ECE = 0.048) succeeds (3 detectors, 3 seeds). Third, mechanism decomposition reveals that minimum intervention—applying the smallest exponent that meets a quality floor—outperforms unconstrained maximization, which over-sharpens and degrades training stability.

As a practical corollary, these findings yield a dual-constrained adaptive b^* that eliminates per-model grid search, achieving perplexity lower than uniform across all seeds at $\rho \geq 0.4$ with calibrated detectors (Δ=−0.020 at ρ=0.4, directionally consistent across all seeds, bootstrap CI excludes zero; Δ=−0.032 at ρ=0.6, uncorrected p=0.010). The method transitions from harmful (ρ=0.2) to beneficial ($\rho \geq 0.4$), with a sharp boundary in $[0.3, 0.4]$; the benefit diminishes at 1.7B, and sample size (N=3) limits generalizability.

1 Introduction

The detect-and-reweight paradigm (Drayson et al., 2025) has emerged as a response to synthetic contamination in web corpora, now estimated at 20–60% of crawled data (Cheng et al., 2025). Training on such data risks model collapse (Shumailov et al., 2024; Dohmatob et al., 2025), motivating methods that downweight likely-synthetic samples via $w(x) = (1 - q(x))^b$, where $q(x)$ is a detector's synthetic probability and b is a sharpening exponent. An alternative, Detect and Remove (D&R), hard-filters samples above a threshold (Feng et al., 2025).

Despite growing adoption, the paradigm's effectiveness conditions are poorly characterized. Does D&R genuinely outperform soft reweighting, or do published comparisons confound filtering quality with multi-epoch exposure? What detector properties are necessary for effective reweighting? How does the reweighting mechanism behave across contamination levels, and can b be set without per-model grid search?

We investigate these questions through controlled experiments on Qwen3-0.6B continued pretraining (132M FineWeb tokens, ρ=0.4, 3 seeds), with extensions to $\rho \in \{0.2, 0.6\}$ and 1.7B parameters (§7 discusses scope). To ground the analysis of exponent selection, we introduce a dual-constrained formulation: find the minimum b such that $\alpha_{\text{eff}}(b) \geq \tau$ (quality floor, motivated by convergence theory (Wang et al., 2026)), subject to $\text{ESS}(b) \geq n_{\min}$ (stability ceiling) (Kish, 1965; Vehtari et al., 2024).

Our analysis yields three findings:

- **Epoch confound in D&R evaluation.** The bulk of D&R's PPL advantage over adaptive reweighting (>95%) is attributable to multi-epoch exposure, not filtering quality; epoch-matched controls (N=3) reduce the gap from 0.396 to Δ=0.017 (Table 1).

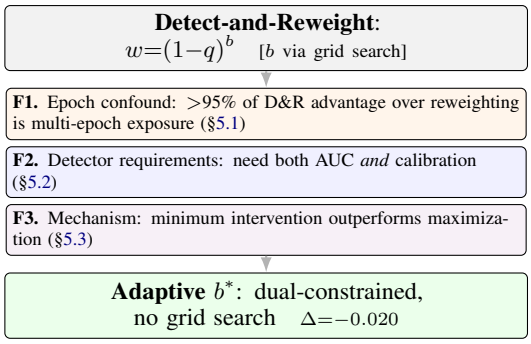


Figure 1: Analysis overview. We identify three effectiveness conditions for detect-and-reweight, from which an adaptive b^* follows as corollary (§3).

- **Detector requirements for effective reweighting.** Both sufficient discrimination (AUC) and good calibration (ECE) are necessary; a high-AUC but poorly calibrated detector degrades performance below uniform (3 detectors, 3 seeds; §5.2).
- **Minimum intervention outperforms maximization.** An α_{eff} quality constraint binds in 50.9% of batches at $5.8\times$ lower sharpening exponents than an ESS-only alternative (§5.3); at ρ=0.6, ESS-only collapses toward uniform while the dual constraint maintains its advantage.

As a practical corollary, we derive a dual-constrained adaptive b^* that eliminates per-model grid search ($\Delta=-0.020$ at ρ=0.4, -0.032 at ρ=0.6; Table 2), with transferable defaults (τ=0.7, $n_{\min}=\lfloor B/2 \rfloor$). The method fails at ρ=0.2; at 1.7B it shows no degradation but the benefit diminishes.

2 Related Work

Synthetic Contamination and Model Collapse. Training on recursively generated data leads to irreversible quality degradation (*model collapse*) (Shumailov et al., 2024). Alemohammad et al. (2024) formalize this as Model Autophagy Disorder and show that quality or diversity inevitably degrades without fresh real data, even when original training data is retained. More critically, Dohmatob et al. (2025) prove under scaling-law regimes that even a vanishingly small synthetic fraction (e.g., $1/1000$) suffices to trigger collapse, ruling out data scaling as a remedy; Dohmatob et al. (2024) characterize this as a change of scaling laws. Gerstgrasser et al. (2024) show that accumulating real data alongside synthetic data can break the recursion, though at substantial collection cost. Kang et al. (2025) systematically study scaling laws for synthetic data in pretraining, finding that rephrased synthetic data mixed with natural text improves data efficiency but textbook-style data degrades loss on many domains. Cheng et al. (2025) survey the broader contamination landscape and confirm that synthetic content now constitutes an estimated 20–60% of web corpora. As language models are deployed at scale for content generation (Radford et al., 2019; Brown et al., 2020; OpenAI, 2023; Dubey et al., 2024), these results motivate *contamination-aware* training strategies that explicitly account for synthetic provenance rather than treating all tokens equally.

Detection and Filtering of Machine-Generated Text. Given the severity of contamination, a natural response is to detect and remove synthetic content. DetectGPT (Mitchell et al., 2023) performs zero-shot detection via probability curvature; Fast-DetectGPT (Bao et al., 2024) accelerates this with conditional probability curvature; Binoculars (Hans et al., 2024) uses cross-perplexity ratios between two models; Min-k% Prob (Shi et al., 2024) uses the average log-probability of the lowest-k% tokens. All produce continuous confidence scores suitable as input to downstream weighting, including ours. Kirchenbauer et al. (2023) propose watermarking as a complementary provenance mechanism. Wu et al. (2025) survey the LLM-generated text detection landscape—the upstream task on which our provenance detector relies—covering statistical, neural, and watermark-based approaches alongside open challenges such as out-of-distribution robustness. On the mitigation side, Feng et al. (2025) prove that even an imperfect verifier prevents collapse, but apply hard binary filtering that discards borderline samples; Yi et al. (2025) extend this with convergence guarantees for verified synthetic data. Zibakhsh Shabgahi et al. (2025) propose truncated cross-entropy loss to implicitly discount high-confidence (likely synthetic) tokens during training. Hard filtering requires a precision–recall threshold; soft reweighting avoids this trade-off by assigning continuous weights.

Most directly related to our work, Drayson et al. (2025) propose $w_j = (1-q_j)^b$ for collapse prevention via detect-and-reweight, where q_j is the detector's synthetic probability. We adopt the same weighting formula but differ in several fundamental ways. **(i)** Drayson et al. select b via validation-set grid search and fix it throughout training; different model scales require different values (b=10 for $\leq$360M, b=1 for 1.7B), so the choice is *per-*

model and expensive. Our b^* is derived per-batch from dual constraints ($\alpha_{\text{eff}} \geq \tau$, ESS $\geq n_{\min}$) at zero validation cost. **(ii)** Their SIR resampling discretizes the weight distribution; we apply direct loss reweighting with continuous gradients. **(iii)** They employ a ModernBERT-150M detector (AUC = 0.986) on WikiText-2 (∼2M tokens); we show that a lightweight TF-IDF+LR detector (AUC = 0.939) suffices on FineWeb-scale data (132M+ tokens), and both calibration and sufficient discrimination appear necessary for effective reweighting.

Domain-Level Data Mixing. A parallel line of work optimizes data composition without targeting synthetic contamination. DoReMi (Xie et al., 2023a) learns mixture weights via Group DRO (Sagawa et al., 2020) with a proxy model, DSIR (Xie et al., 2023b) performs importance resampling with n-gram features, and Ye et al. (2025) discover predictable functional forms relating mixing ratios to performance. AIOLI (Chen et al., 2025) unifies these into an online adaptive framework, while Yang et al. (2026) train an RL agent for domain-weight selection. DataComp-LM (Gadre et al., 2024) and OLMo (Groeneveld et al., 2024) provide standardized benchmarks and open infrastructure for data curation at scale. ToReMi (Zhu et al., 2025) extends domain-level mixing by clustering training data into topics and dynamically adjusting category-level weights based on aggregate learning signals. Continued pretraining (Gururangan et al., 2020) adapts models to specific domains, a setting where contamination-aware methods are directly applicable. These domain-level approaches do not address instance-level contamination within a domain.

Instance-Level Data Selection and Reweighting. Deduplication at the corpus level (Lee et al., 2022) is a prerequisite; data-constrained regimes (Muennighoff et al., 2024) and data quality filtering (Longpre et al., 2024) further shape training distributions. SemDeDup (Abbas et al., 2023) deduplicates via embedding similarity, SoftDedup (He et al., 2024) downweights high-frequency samples via n-gram commonness, and QuRating (Wettig et al., 2024) selects high-quality instances via learned quality scores. Curriculum learning (Bengio et al., 2009) and focal loss (Lin et al., 2017) upweight examples by difficulty, Katharopoulos and Fleuret (2018) accelerate training via importance sampling of high-loss examples, Mindermann et al. (2022) prioritize samples by reducible loss, Ren et al. (2018) meta-learn per-sample weights for noisy-label robustness, and Sow et al. (2025) derive sample weights from per-sample loss with convergence guarantees. LESS (Xia et al., 2024) selects influential data for instruction tuning, Chen et al. (2024) apply hard sample reweighting during LLM continual pretraining, and Kuo et al. (2024) show that differentiated weighting of LLM-generated data improves downstream classification. Thakkar et al. (2023) use self-influence scores to reweight pretraining samples, promoting novelty and penalizing redundancy; their PRESENCE method operates on intrinsic training signals rather than provenance. Pruthi et al. (2020) trace gradient descent to estimate per-example training data influence, enabling post-hoc attribution but at prohibitive cost for pretraining-scale data. Our reweighting differs from all of these in being driven by *provenance* (detector scores estimating human vs. synthetic origin) rather than loss, frequency, quality, or domain membership.

Theoretical Foundations. He et al. (2025) derive the optimal real-data weight under known mixture proportions, finding the golden-ratio reciprocal (≈0.62) optimal when both sources are equally sized. Their result requires the contamination ratio k, unavailable in web-scale settings; we use it as an oracle baseline. Wang et al. (2026) prove that contaminated recursive training converges at a rate governed by the effective real-data fraction α; this result provides the theoretical motivation for our quality threshold, since $\alpha_{\text{eff}}(b) \geq \tau$ guarantees a convergence-rate lower bound. We operationalize this without the known-proportion assumption, estimating α_{eff} from batch-level statistics online. For the stability constraint, we build on Kish's effective sample size (Kish, 1965) and the Pareto-smoothed importance sampling framework of Vehtari et al. (2024). Score calibration (Guo et al., 2017) is relevant because the adaptive method operates on q-scores: miscalibration systematically biases α_{eff}. Kimura and Hino (2024) provide a broader treatment of importance weighting under distribution shift, within which our provenance-based reweighting can be viewed as a special case of covariate-shift correction where the covariate is synthetic origin probability.

3 Method

Let $\mathcal{D} = \mathcal{D}_{\text{real}} \cup \mathcal{D}_{\text{synth}}$ denote a pretraining corpus with token-level contamination ratio $\rho =$

$|\mathcal{D}_{\text{synth}}|/|\mathcal{D}|$, assumed unknown (Cheng et al., 2025). A detector assigns each sample x a synthetic probability $q(x) \in [0, 1]$; following Drayson et al. (2025), we weight each sample as $w(x) = (1 - q(x))^b$, where $b \geq 0$ is a sharpening exponent (b=0 recovers uniform training). Rather than selecting b via grid search (Drayson et al., 2025), we derive b per-batch from the batch's q-score distribution using two opposing constraints.

Given a batch $\mathcal{B} = \{x_1, \ldots, x_B\}$ with scores $q_i = q(x_i)$ and weights $w_i(b) = (1 - q_i)^b$:

Quality floor. Let $r_i = \mathbf{1}[q_i \leq 0.5]$ indicate likely-real samples. The *effective real proportion* $\alpha_{\text{eff}}(b) = \sum_i w_i(b) r_i / \sum_i w_i(b)$ measures weight allocated to real samples. We require $\alpha_{\text{eff}}(b) \geq \tau$, yielding a lower bound b_τ on the exponent, motivated by convergence theory under contamination (Wang et al., 2026). α_{eff} correlates with the true batch-level real fraction at Pearson $r \approx 0.70$ (Appendix K); limitations are discussed in §6.

Stability ceiling. Kish's effective sample size $\text{ESS}(b) = (\sum_i w_i)^2 / \sum_i w_i^2$ (Kish, 1965) decreases monotonically in b (Appendix A). We require $\text{ESS}(b) \geq n_{\min}$, yielding an upper bound b_{ESS}.

Adaptive b^*. Since α_{eff} is non-decreasing and ESS is decreasing in b, the two constraints define an interval $[b_\tau, b_{\text{ESS}}]$. The optimal exponent implements *minimum intervention*:

$$b^* = \begin{cases} 0 & \text{if } \alpha_{\text{eff}}(0) \geq \tau, \\ b_\tau & \text{if } b_\tau \leq b_{\text{ESS}}, \\ b_{\text{ESS}} & \text{otherwise (stability precedence)}, \end{cases} \tag{1}$$

applying the smallest b that achieves the quality floor. Each batch falls into one of three regimes—clean, α_{eff}-binding, or ESS-binding—whose frequencies we analyze in §5.3. Given b^*, normalized weights $\pi_i = w_i(b^*) / \sum_j w_j(b^*)$ define the weighted loss $\mathcal{L} = \sum_i \pi_i \ell_i$; b^* is found via bisection (Algorithm 1, Appendix B). Defaults: τ=0.7, $n_{\min}$=$\lfloor B/2 \rfloor$; sensitivity in §5.3.

4 Experimental Setup

Training Data. We sample a 132M-token subset of FineWeb (Penedo et al., 2024) as the real-data source, comparable to typical domain-adaptive continued pretraining budgets (Gururangan et al., 2020). To simulate web-scale synthetic contamination, we generate continuations with Qwen3-0.6B (Yang et al., 2025), the same architecture used for continued pretraining, and inject them at token-level ratio $\rho = 0.4$ via uniform random mixing.[1] This self-contamination design means generator and trainee share the same distribution; implications for detection difficulty are discussed in §7. Our self-contamination design provides a controlled testbed with known ground-truth labels but limits ecological validity; cross-model transfer results (Appendix O) are mixed, and we discuss this limitation in §7.

Detector. We train a TF-IDF + Logistic Regression classifier on a held-out labeled split (AUC = 0.94, calibrated ECE = 0.05 after temperature scaling with T=0.204). The resulting q-scores have mean 0.60, standard deviation 0.37, and median 0.71, with 46.3% of samples in the uncertain range $[0.2, 0.8]$. We use this lightweight detector rather than neural alternatives (e.g., ModernBERT at AUC $\geq$ 0.98 in Drayson et al. 2025) to test whether score *calibration* may matter as much as raw discrimination for reweighting. To evaluate detector sensitivity, we additionally test two generative baselines: a GPT-2-based perplexity scorer (AUC = 0.91) and an N-gram language model perplexity scorer (AUC = 0.73); results are reported in §5.2.

Training. We continue pretraining Qwen3-0.6B from its pretrained weights for 6,103 steps (batch size 4, gradient accumulation 4, effective batch 16) with AdamW, linear schedule with warmup (peak lr = 5×10^{-5}), max sequence length 2048, and bf16 precision on a single NVIDIA RTX PRO 6000 (96 GB). All main comparisons use three random seeds (42, 123, 456).

Evaluation. We report held-out perplexity (PPL) on 5,000 documents from a FineWeb holdout set ($\sim$4M tokens) containing only human-authored text, via sliding-window evaluation (stride 1024). All evaluations are post-hoc with a unified pipeline; training involves no inline evaluation.

Methods. We compare six approaches:

- **Uniform**: $w_j = 1$ for all j (no contamination handling).
- **Grid-b**: fixed $b \in \{0.1, 0.2, 0.3, 0.5, 1.0, 1.5, 2.0, 3.0, 5.0\}$

[1] ρ denotes the token-level contamination ratio. Because synthetic documents are shorter on average (304 vs. 560 words), the document-level synthetic fraction is 55.6% at ρ=0.4.

(nine candidates), best selected post-hoc by held-out PPL. Represents validation-based selection (Drayson et al., 2025).

- **ESS-only**: $b^* = \max\{b : \mathrm{ESS}(b) \geq n_{\min}\}$, the maximum-intervention variant retaining only the stability constraint, ablating α_{eff}.
- **Golden Ratio**: $w = 1/\varphi \approx 0.618$ for synthetic ($q_j > 0.5$), $w = 1$ otherwise (He et al., 2025). Requires known provenance labels (oracle).
- **Adaptive** (ours): per-batch b^* from the dual-constraint formulation (§3), with $\tau = 0.7$ (default; τ=0.8 comparable, §5.3) and $n_{\min} = 2$ (half the micro-batch size).
- **Detect & Remove**: hard filtering that discards samples with $q_j > 0.5$ and trains uniformly on the remainder. Requires a calibrated detection threshold; threshold sensitivity is analyzed in §5.1.

5 Results

5.1 Main Results

Table 2 reports held-out perplexity at $\rho = 0.4$ across three seeds. Continued pretraining reduces PPL from 22.970 to the 17.3–17.8 range; method comparisons operate within a narrow residual.

Adaptive b^* achieves 3-seed mean PPL 17.674, compared to 17.694 for uniform (Δ=-0.020, directionally consistent across all seeds) and 17.620 for grid b=0.3 (3-seed mean; gap +0.054). A paired t-test yields t=-7.1, p=0.019; after Bonferroni–Holm correction for nine family-wise comparisons, p_{BH}=0.078, which does not reach conventional significance, though the 95% BCa bootstrap CI $[-0.028, -0.012]$ excludes zero. The practical value is eliminating b grid search (nine full training runs (Drayson et al., 2025), ∼16 GPU-hours at 0.6B scale, scaling linearly with model size), though grid search yields a larger improvement. The grid-best b=0.3 is a post-hoc oracle from nine candidates with 4.7× higher cross-seed variance than adaptive (σ=0.033 vs. 0.007); the noisy PPL plateau at $b \in [0.1, 0.5]$ (Figure 2) makes validation-based selection unreliable. At the nearest fixed point (b=1.0), PPL equals uniform (17.692 vs. 17.694): per-batch adaptation is essential.

The ablation hierarchy at ρ=0.4 (Adaptive < Uniform < ESS-only < Random, Table 2) reveals two requirements for effective reweighting: detector signal and minimum intervention. Random weights $\mathcal{U}(0, 2)$ (17.802) confirm that detector q-scores carry genuine provenance signal; uninformed reweighting only increases gradient variance. ESS-only (mean 17.737±0.021) uses the same q-scores but maximizes b up to the stability ceiling, concentrating weights on a few high-confidence samples; it degrades PPL by 0.043 relative to uniform. Only the dual-constrained adaptive method, which limits b to the minimum needed, improves in-distribution PPL over uniform (mean Δ=-0.020, directionally consistent across all seeds). Figure 2 contextualizes both results: adaptive b^* operates near the curve minimum while ESS-only b^* (mean 3.5–5.9) falls in the degradation zone.

Hard filtering and the epoch confound. Detect & Remove at its best threshold (q>0.3) achieves 3-seed mean PPL 17.250 ± 0.001 at full training length, apparently outperforming all reweighting methods. However, filtering retains only 31.5% of tokens, so the model sees surviving data ∼3.2 *content epochs* versus one for reweighting. An epoch-matched control (D&R at q>0.5, 2,533 steps, ∼1.0 content epochs) yields PPL 17.657 ± 0.005 vs. 17.674 ± 0.007 for adaptive b^* (Δ=0.017); the bulk of D&R's full-length advantage (0.396 PPL) is attributable to multi-epoch exposure (Table 1). The raw threshold–PPL curve is U-shaped (q>0.3: 17.25, q>0.9: 17.42), but epoch-matched comparisons eliminate the U-shape, revealing monotonically decreasing PPL that saturates at $q \geq 0.5$ (Figure 5, Appendix H): the apparent optimum at q>0.3 reflects maximally compensating multi-epoch exposure, not filtering quality.

LR-controlled verification. To rule out the LR confound (epoch-matched D&R inherits the full-length schedule's LR trajectory), we retrain D&R (q>0.5) and uniform with independent linear schedules (2,533 steps each, LR → 0). D&R achieves 17.651, matching the shared-schedule result (17.657, Δ=0.006); uniform achieves 18.062, confirming the epoch-exposure advantage (Δ=-0.411) persists under controlled LR.

5.2 Robustness and Boundary Conditions

Across three seeds at $\rho = 0.6$, adaptive b^* (17.731 ± 0.004) outperforms uniform on in-distribution PPL (17.763 ± 0.008, Δ=-0.032, paired t=-9.95, p=0.010[2]), exceeding the ρ=0.4

[2]Computed from 3dp-rounded per-seed values (Table 10) for reproducibility. Under full-family Bonferroni–Holm correction (k=9), p_{BH}=0.064; directionally consistent across all

Method	Steps	Epochs	PPL ↓
D&R full (q>0.5)	6,103	∼2.4	17.278 ± .007
D&R epoch-matched	2,533	∼1.0	17.657 ± .005
Adaptive b^*	6,103	∼1.0	17.674 ± .007

Table 1: Epoch-matched comparison (ρ=0.4, 3 seeds). D&R's 0.396 PPL advantage over adaptive shrinks to Δ=0.017 when content epochs are equalized. Note: epoch-matched D&R shares the full-length LR schedule; an independent-schedule replication (D&R 17.651, uniform 18.062) matches the shared-schedule result (Δ=0.006), confirming the epoch-exposure advantage persists under controlled LR.

Method	Seed 42	Seed 123	Seed 456	Mean
Pretrained ckpt	22.970	–	–	–
Uniform	17.694	17.686	17.702	17.694 ± .008
Grid (b=0.3)‡	**17.640**	**17.579**	**17.640**	**17.620** ± .033
Adaptive b^* (ours)	17.680	17.666	17.677	17.674 ± .007
ESS-only	17.730	17.720	17.760	17.737 ± .021
Random $\mathcal{U}(0,2)$	17.830	17.800	17.777	17.802 ± .027
Golden Ratio†	17.654	–	–	–
Detect & Remove§	**17.250**	**17.250**	**17.251**	**17.250** ± .001

Table 2: Held-out perplexity (↓) on human-authored FineWeb at ρ=0.4. All reweighting methods run across three seeds. Grid-best (b=0.3) selected post-hoc from nine candidates (Appendix C). Best per column in **bold**. † Oracle requiring known ρ (He et al., 2025); seed 42 only. ‡ Post-hoc oracle from nine candidates. § Hard filtering at q>0.3 (post-hoc best threshold); ∼3.2 content epochs (see Table 1). Per-seed values rounded to three decimal places; means and standard deviations computed from full precision. Adaptive uses τ=0.7 (default; τ=0.8 comparable, §5.3). All adaptive runs trained from scratch; see Appendix P for an early checkpoint artifact that does not affect these results.

improvement (Δ=−0.020). The ablation hierarchy is preserved (Grid 17.634[3] < Adaptive 17.731 < Uniform 17.763 ≈ ESS-only 17.763 < Random 17.861), but the ESS-only/uniform gap collapses to effectively zero (three-seed means both 17.763) as ESS binding rises to ∼49%. Random remains substantially worse than uniform (gap 0.098, σ=0.025): detector signal is load-bearing even when the stability constraint dominates; the adaptive method's continued advantage is driven by the α_{eff} quality constraint, which operates independently of the ESS ceiling. Three-seed paired comparison confirms τ=0.7 outperforms τ=0.8 on both holdout and OOD metrics (§5.3).

seeds.

[3]Single seed; full per-seed results in Appendix J.

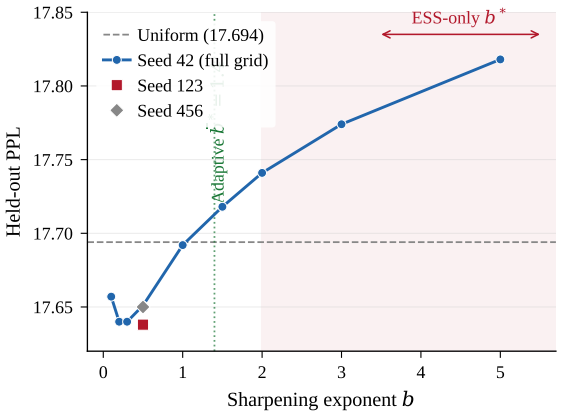


Figure 2: Held-out PPL as a function of fixed sharpening exponent b (grid search, ρ=0.4). Seed 42 provides the full nine-point sweep (circles); cross-seed markers at b=0.5 (triangles) confirm the plateau region. The shaded region marks $b \geq 2$ where PPL degrades past the uniform baseline; ESS-only b^* (mean 3.5–5.9 across batches) falls in this degradation zone, while adaptive b^* (mean 1.4) operates near the curve minimum.

Scale. At 1.7B parameters (Qwen3-1.7B, ρ=0.4, τ=0.8, 3 seeds), the method scales without degradation but the benefit diminishes. Adaptive b^* achieves holdout PPL 13.26 vs. uniform 13.28 (Δ=−0.013); OOD results are partially consistent (OWT Δ=−0.008, Wiki Δ=+0.013) and downstream is unchanged (Δ≈+0.001), consistent with larger models being more robust to 0.6B-generated contamination.

Out-of-distribution evaluation. To test whether the in-distribution improvement reflects distribution matching, we evaluate all three from-scratch seeds on OpenWebText and Wikipedia. At ρ=0.4, adaptive reweighting improves 3-seed mean PPL on all three evaluation domains: FineWeb Δ=−0.020, OpenWebText Δ=−0.028, Wikipedia Δ=−0.029 (downstream accuracy identical at 0.5414). OOD improvements exceed the in-distribution gain, indicating genuine quality improvement rather than distribution matching toward the training mixture (τ=0.8's OOD effect is smaller; τ=0.7's advantage reflects active OOD improvement, §5.3).

Calibration, not just discrimination. Replacing TF-IDF+LR with generative perplexity detectors reverses the advantage across three seeds (GPT-2: AUC = 0.91, ECE = 0.131, Δ=+0.098; N-gram: AUC = 0.73, ECE = 0.069, Δ=+0.129; Appendix L). The TF-IDF vs. GPT-2 contrast (com-

parable AUC, 2.7× ECE gap) isolates calibration as one necessary condition; N-gram's better ECE (0.069) does not compensate for insufficient AUC (0.73), confirming that both discrimination and calibration are required.

5.3 Hyperparameter Sensitivity

Quality threshold τ. τ=0.7 outperforms τ=0.8 on holdout at both contamination levels (ρ=0.4: 17.674 vs. 17.683; ρ=0.6: paired t=−4.70, p<0.05; §5.2) and achieves larger OOD gains (ρ=0.4: Δ=−0.028/−0.029 vs. −0.005/−0.009; Table 6, Appendix G). $\tau \in [0.5, 0.8]$ all improve holdout over uniform, while τ>0.8 degrades (Appendix D).

ESS floor $n_{\min}$. The default $n_{\min}$=2 ($n_{\min}/B$=0.5) is optimal; extremes degrade ($n_{\min}$=1: PPL 17.789; $n_{\min}$=B: recovers uniform). Under $n_{\min}$=$\lfloor B/2 \rfloor$ coupling, the adaptive advantage monotonically increases across $B \in \{4, 8, 16\}$: Δ=−0.020±0.008 (3 seeds), −0.050 (2-seed mean), and −0.068±0.005 (3 seeds), corresponding to 1.0×, 2.5×, and 3.4× amplification (p<0.01 at B=16). The gap to grid-best narrows from 0.040 (B=4) to 0.009 (B=16, seed 42).

Constraint regime. Over 24,412 micro-batches (seed 42), 5.8% are clean ($\alpha_{\text{eff}}(0) \geq \tau$, b^*=0), 50.9% are α_{eff}-binding (quality constraint dominates), and 43.3% are ESS-binding (stability constraint dominates); the distribution is stable across seeds (CV <2%; Appendix F). The α_{eff}-binding majority indicates that detector quality signal—not training stability—is the primary bottleneck, reinforcing the calibration requirement (§5.2).

6 Discussion

The epoch confound in D&R evaluation. The bulk of D&R's reported PPL advantage over adaptive reweighting (>95%) is attributable to multi-epoch exposure rather than filtering quality (§5.1). An independent-LR replication (D&R 17.651 vs. shared-schedule 17.657, Δ=0.006) confirms that this finding is robust to LR schedule choice. D&R evaluations that do not control for content epochs systematically overstate filtering effectiveness; the residual benefit (Δ=0.017 vs. adaptive b^*, directionally consistent across 3 seeds) falls within cross-seed variance. A permuted-score control (shuffled detector scores; 3-seed mean 17.951±0.007, +0.26 vs. uniform) confirms that detector signal is load-bearing: uninformed reweighting only degrades.

Condition	Req'd?	Evidence	Δ PPL
Detector AUC ≥ 0.9	Yes	N-gram (0.73) degrades	+0.129
Detector ECE ≤ 0.05	Yes	GPT-2 (0.131) degrades	+0.098
Contamination $\rho \geq 0.4$	Yes	ρ=0.2 degrades	+0.046
Generator ≠ trainee	Rec.	1.7B: Δ=−0.013, OOD mixed	—

Table 3: Decision summary for practitioners. Δ values are relative to uniform (positive = worse). When any required condition is violated, uniform is the safer default.

When uniform beats reweighting. Beyond the evaluation confound, our results identify three operational conditions for effective reweighting, each of which, when violated, makes uniform training preferable. First, *detector quality*: the detector must provide both sufficient discrimination (AUC) and good calibration (ECE). A high-AUC but poorly calibrated detector (GPT-2, ECE = 0.131) degrades PPL by +0.098 below uniform; even a well-calibrated detector with insufficient discrimination (N-gram, AUC = 0.73) degrades by +0.129 (§5.2). Second, *contamination level*: at ρ=0.2, the detector's signal-to-noise ratio is too low and the method actively intervenes on unreliable signal (α_{eff}-binding at 60.1% despite mean q=0.393), degrading PPL by +0.046; the method transitions from harmful to beneficial with a sharp sign change in $[0.3, 0.4]$ (Figure 6, Appendix Q). Third, *scale*: at 1.7B the benefit diminishes (Δ=−0.013) with OOD partially inconsistent, suggesting that larger models may be inherently more robust to contamination generated by smaller models. In all three failure cases, the adaptive method's dual constraint limits degradation relative to unconstrained reweighting (ESS-only), but uniform remains the safer default when conditions are uncertain (Table 3).

Downstream tasks do not reflect PPL improvements. The boundary conditions above frame the practical significance of the observed improvements. Despite consistent holdout PPL improvements across seeds (Δ=−0.020 at ρ=0.4), downstream benchmark accuracy is identical between adaptive and uniform training (Table 4). Across three seeds, the mean accuracy difference is ≤0.001 with mixed directionality; no individual task shows a consistent advantage. This null result is informative: it suggests that the PPL im-

	Adaptive	Uniform
Mean accuracy (3 seeds)	0.5414	0.5414
Mean $\lvert\Delta\rvert$ per task	≤ 0.01	
Direction	mixed across seeds	

Table 4: Downstream benchmark summary (ρ=0.4, 5 tasks, 3 seeds; per-task details in Appendix I). All differences within standard error; reweighting PPL gains do not transfer to downstream accuracy at this scale.

provements from contamination-aware reweighting operate in a regime that standard benchmarks cannot resolve. Quantitatively, Δ=-0.020 PPL corresponds to $\sim$0.1% relative improvement; on 5-choice tasks with $\sim$1,000 evaluation items, each question contributes $\sim$0.001 to accuracy, so the expected accuracy change from a 0.1% PPL shift is far below per-task measurement noise ($\sigma \approx 0.01$). More broadly, the relationship between pretraining perplexity and downstream accuracy is nonlinear and scale-dependent (Tay et al., 2023): small PPL gains at 0.6B may not cross the threshold where downstream metrics begin to respond, consistent with findings in data mixing (Ye et al., 2025) where optimized mixtures similarly fail to transfer at smaller scales.

Effect size in context. These results raise the question of whether the observed improvements justify the method's overhead. The in-distribution PPL improvement from adaptive reweighting (Δ=-0.020 at ρ=0.4) operates within a narrow residual: continued pretraining absorbs $\sim$5 PPL points of easy gains (22.97 $\rightarrow$ 17.69), leaving method comparisons to operate within a $\sim$0.07-point window between uniform and grid-best (0.18 across all methods including random). After accounting for the epoch confound, the true value added by detect-and-reweight is substantially smaller than published results suggest. This finding is itself valuable: the detect-and-reweight paradigm's practical impact at current scales is modest, and practitioners should weigh the method's computational cost (detector training, per-batch b^* computation) against the expected gain. At ρ=0.6 the effect roughly doubles (Δ=-0.032), suggesting that the method's value may increase as web contamination grows—but this extrapolation is speculative.

Limitations of α_{eff} as a proxy. The α_{eff} diagnostic correlates with the true batch-level real fraction at Pearson $r \approx 0.70$, explaining $\sim$49% of variance (Appendix K). The proxy correlation is stable across seeds: $r(\alpha_{\text{eff}}, \bar{q}) = 0.80 \pm 0.001$ (cross-seed spread <0.003, Spearman $\rho \geq 0.87$; Table 12). Note that $r(\alpha_{\text{eff}}, \bar{q}) = 0.80$ measures proxy-to-proxy (detector score) correlation, while $r(\alpha_{\text{eff}}, \text{true real fraction}) = 0.706$ (Table 11) measures proxy-to-ground-truth; the gap reflects thresholding noise in r_i, and the key information from the per-seed analysis is the cross-seed stability, not the absolute value. The remaining variance arises from two sources: the binary indicator $r_i = \mathbf{1}[q_i \leq 0.5]$ discards calibration information within each half of the q-score range, and small batch sizes (B=4) amplify sampling noise. Quantitative decomposition attributes $\sim$70% of unexplained variance to detector noise (TF-IDF accuracy drops to $\sim$55% in the boundary region $q \in [0.4, 0.6]$) and $\sim$30% to the adaptive b^* transform, which intentionally compresses α_{eff} toward τ via the ESS $\geq n_{\min}$ constraint. Scaling from B=4 to B=16 reduces per-batch α_{eff} variance by 10.25$\times$ but leaves R^2 essentially unchanged across batch sizes ($\approx$0.50 at both B=4 and B=16), confirming that the practical ceiling is set by detector calibration quality rather than batch size. Increasing B nonetheless improves the adaptive advantage monotonically (1.0$\times$ at B=4, 2.5$\times$ at B=8 2-seed mean, 3.4$\times$ at B=16 3 seeds; §5.3), but with diminishing marginal gains (2.5$\times$ for the first doubling, 1.36$\times$ for the second). Over the tested range, the effect scales as $\Delta \propto B^{\alpha}$ with $\alpha \approx 0.89$, suggesting diminishing but substantial marginal gains at larger batch sizes. However, the r=0.70 correlation sets a ceiling on how precisely the method can target contamination: a better proxy (e.g., using soft q-scores directly rather than thresholded indicators) might improve the method's effective range, particularly at ρ=0.2 where the current proxy fails.

7 Conclusion

We identify three effectiveness conditions for detect-and-reweight: (1) D&R evaluations confound filtering with multi-epoch exposure (>95% of the advantage over adaptive reweighting); (2) detectors require both discrimination and calibration; (3) minimum intervention outperforms unconstrained maximization. A dual-constrained adaptive b^* (τ=0.7, $n_{\min}$=$\lfloor B/2 \rfloor$) improves over uniform at $\rho \geq 0.4$ (directionally consistent across all seeds; ρ=0.6: uncorrected p=0.010, p_{BH}=0.064 with k=9) without tuning; validated on Qwen3

≤1.7B, where the benefit diminishes. Batch-size amplification is monotonic across $B \in \{4, 8, 16\}$ ($1.0\times$ to $3.4\times$; $p<0.01$ at $B=16$) with diminishing marginal gains, suggesting favorable scaling with larger micro-batches typical of production training; the key bottleneck is extending detector calibration to cross-model contamination scenarios.

The epoch confound finding has implications beyond our experiments: any D&R evaluation that compares at equal training steps without controlling for content epochs risks overstating filtering effectiveness, and published results in this area should be interpreted with this confound in mind. For practitioners, the decision framework (Table 3) distills our analysis into actionable guidance: provenance-aware reweighting is worthwhile only when contamination is substantial ($\rho \geq 0.4$) and the detector is well-calibrated (ECE ≤ 0.05); otherwise, uniform training remains the safer default. Key directions for future work include cross-model contamination scenarios (where the generator differs from the trainee), better proxies for batch quality that exploit soft q-scores rather than binary indicators, and larger-scale verification to test whether the batch-size amplification trend continues at production training scales. Code and data will be released upon acceptance.

Limitations

All experiments use Qwen3 (0.6B and 1.7B) on 132–200M tokens of continued pretraining with self-generated contamination at $\rho \in \{0.2, 0.3, 0.4, 0.6\}$; generalization to other model families, full pretraining scale (>100B tokens), or realistic multi-source web contamination is untested. The method improves over uniform only at $\rho \geq 0.4$ with a calibrated detector; at $\rho=0.2$ it degrades, and a preliminary Llama-3.2-3B cross-model test showed mixed results (Appendix O), suggesting the effective range is narrower than the paradigm's ambition. Three detectors from two families (discriminative and generative) were tested; fine-tuned neural classifiers (e.g., RoBERTa-based) and zero-shot methods (e.g., DetectGPT) remain untested, so the calibration finding may not generalize across detector architectures. Paired comparisons use $N=3$ seeds (2 degrees of freedom), limiting statistical power; results are directionally consistent but the sample size precludes strong generalizability claims. After family-wise correction for nine comparisons, the $\rho=0.4$ adaptive-vs-uniform difference does not reach conventional significance ($p_{\mathrm{BH}}=0.078$), though the effect is directionally consistent across all seeds and the bootstrap CI excludes zero; the $\rho=0.6$ comparison also falls short of conventional significance under full-family correction ($p_{\mathrm{BH}}=0.064$, $k=9$), though the uncorrected $p=0.010$ and dose-response pattern across ρ levels support a real effect. The self-contamination setup provides known ground-truth labels but represents an easier detection scenario than cross-model contamination, where generator–corpus distribution mismatch may reduce detector accuracy.

Ethics Statement

All experiments use publicly available datasets (FineWeb) and open-weight models; no human subjects are involved. Code and data will be released upon acceptance.

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A Proofs

Proposition 1 (α_{eff} monotonicity)**.** *$\alpha_{\text{eff}}(b)$ is non-decreasing in b, and strictly increasing whenever the batch contains at least one sample with $q_i \in (0, 1)$ and r_i is not constant.*

Proof of Proposition 1. Let $w_i(b) = (1-q_i)^b$, $W(b) = \sum_{j=1}^{B} w_j(b)$, and $\pi_i(b) = w_i(b)/W(b)$. Then

$$\alpha_{\text{eff}}(b) = \sum_{i=1}^{B} \pi_i(b)\, r_i,$$

where $r_i \in \{0,1\}$ indicates whether sample i is real. Taking the derivative with respect to b:

$$\frac{d\pi_i}{db} = \pi_i\left(\frac{d\ln w_i}{db} - \frac{d\ln W}{db}\right)$$
$$= \pi_i\Big(\ln(1-q_i) - \textstyle\sum_j \pi_j \ln(1-q_j)\Big).$$

Substituting,

$$\frac{d\alpha_{\text{eff}}}{db} = \sum_i r_i\, \pi_i(\ln(1-q_i) - \mathbb{E}_\pi[\ln(1-q)])$$
$$= \text{Cov}_\pi\big(r_i,\, \ln(1-q_i)\big).$$

Since $r_i = \mathbf{1}[q_i \le 0.5]$, the indicator r_i and $\ln(1-q_i)$ are positively associated by construction, so the covariance is non-negative. □

Proposition 2 (ESS monotonicity)**.** $\text{ESS}(b)$ *is non-increasing in* b*, and strictly decreasing whenever the batch contains at least two samples with distinct* q*-values.*

Proof. Write $\text{ESS}(b) = 1/\sum_i \pi_i(b)^2$, where $\pi_i(b) = w_i(b)/W(b)$. The denominator $\sum_i \pi_i^2$ is the Herfindahl index H. Differentiating H with respect to b:

$$\frac{dH}{db} = 2\sum_i \pi_i^2\big(\ln(1-q_i) - \mathbb{E}_\pi[\ln(1-q)]\big).$$

Since $\pi_i \propto (1-q_i)^b$, larger π_i implies larger $\ln(1-q_i)$; the π^2-weighted sum assigns greater mass to terms where $\ln(1-q_i) > \mathbb{E}_\pi[\ln(1-q)]$, so $dH/db \ge 0$, with strict inequality when at least two q-values differ. Thus H is non-decreasing and $\text{ESS} = 1/H$ is non-increasing. This is a standard property of exponential tilting (Kish, 1965; Owen, 2013). □

B Algorithm Pseudocode

Each bisection requires $\mathcal{O}(B\log(b_{\max}/\varepsilon))$ operations; with B=4, $b_{\max}$=20, ε=0.01, roughly 44 floating-point operations per batch, negligible beside the forward and backward passes. We compute b^* at the micro-batch level (B=4) rather than the effective batch (B=16 after gradient accumulation) because q-score statistics exhibit sufficient variance within micro-batches: Spearman r=0.699 with the true real fraction, and at B=16 the adaptive advantage amplifies 3.4× (Δ=−0.068±0.005; §5.3).

Algorithm 1 Per-batch adaptive b^* selection

Require: Batch scores $\{q_i\}_{i=1}^{B}$, indicators $\{r_i\}$, quality threshold τ, ESS floor $n_{\min}$, search bound $b_{\max}$, tolerance ε
Ensure: Optimal sharpening exponent b^*

```
if α_eff(0) ≥ τ then
    return 0 {clean batch}
end if
b_τ ← BISECT(α_eff(·)=τ, [0, b_max], ε)
if ESS(b_τ) ≥ n_min then
    return b_τ {α_eff-binding}
else
    b_ESS ← BISECT(ESS(·)=n_min, [0, b_max], ε)
    return b_ESS {ESS-binding}
end if
```

C Grid Search Details

Table 5 reports held-out PPL for all grid-b values at $\rho = 0.4$ (seed 42).

b	PPL ↓
0.1	17.657
0.2	17.640
0.3	**17.640**
0.5	17.651
1.0	17.692
1.5	17.718
2.0	17.741
3.0	17.774
5.0	17.818

Table 5: Full grid search results (ρ=0.4, seed 42). $b \in [0.1, 0.5]$ forms a broad plateau (three-seed mean for b=0.1: 17.655 ± 0.009); b=0.3 is the post-hoc best.

D Quality Threshold Sweep

E Batch-Adaptive τ

We tested a per-batch adaptive variant that clamps τ between 0.5 and the fixed threshold based on local $\alpha_{\text{eff}}(0)$, allowing the quality floor to relax in batches with low estimated real fraction. At both contamination levels ($\rho \in \{0.4, 0.6\}$), the adaptive variant did not outperform the fixed threshold (single-seed exploratory at each level), suggesting the quality constraint does not require adaptation to local contamination density. This supports the "two hyperparameters with good defaults" characterization: neither τ nor $n_{\min}$ benefits from per-batch tuning.

τ	Holdout ↓	Δ_H	OWT ↓	Δ_O	Wiki ↓	Δ_W
0.50	17.688	−0.006	–	–	–	–
0.60	17.681	−0.013	–	–	–	–
0.70*	**17.674**$_{\pm.007}$	**−0.020**	**18.347**$_{\pm.003}$	**−0.028**	**15.330**$_{\pm.015}$	**−0.029**
0.80*	17.683$_{\pm.006}$	−0.011	18.370$_{\pm.005}$	−0.005	15.350	−0.009
0.90	17.712	+0.018	–	–	–	–
0.95	17.723	+0.029	–	–	–	–

Table 6: Quality threshold sweep ($n_{\min}$=2, ρ=0.4). Δ relative to uniform baseline (Holdout 17.694, OWT 18.375, Wiki 15.359). *τ=0.7: 3 seeds; τ=0.8: 3 from-scratch seeds (Appendix P). Other values from single-seed exploratory sweep (seed 42, holdout only). τ=0.7 achieves the lowest holdout and largest OOD gains ($|\Delta| \geq 0.028$); τ=0.8's OOD effect is smaller ($|\Delta| \leq 0.009$).

F Constraint Regime Decomposition

Figure 3 visualizes the constraint regime decomposition at two contamination levels.

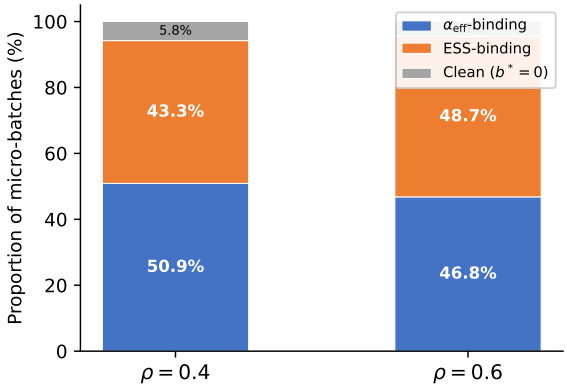


Figure 3: Constraint regime decomposition at two contamination levels (seed 42). At ρ=0.4, 50.9% of batches are α_{eff}-binding and 43.3% ESS-binding, with 5.8% clean (b^*=0); the quality constraint dominates. At ρ=0.6, the constraints are nearly balanced: 48.7% ESS-binding and 46.8% α_{eff}-binding, with 4.5% clean; ESS-binding increases modestly from ρ=0.4 (43.3%→48.7%) as contamination raises the stability pressure.

G τ–ρ Rank Preservation

Figure 4 shows the τ–PPL relationship at two contamination levels.

H Hybrid and Threshold Comparison

Table 7 compares method families across contamination levels. Table 8 shows D&R sensitivity to the detection threshold at both full training length and epoch-matched settings. Raw PPL traces a U-shaped curve with a minimum at q>0.3 (17.250, 3 seeds), but this curve conflates filtering quality with

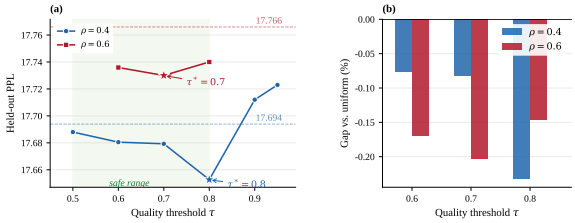


Figure 4: (a) Held-out PPL as a function of quality threshold τ at $\rho \in \{0.4, 0.6\}$. τ=0.7 and τ=0.8 yield comparable holdout PPL at ρ=0.4; at ρ=0.6, τ=0.7 is statistically better (paired t=−4.70, p<0.05, 3 seeds, Δ=−0.005). τ=0.7 also achieves lower OOD perplexity at both levels (ρ=0.6: Δ_{OWT}=−0.014, Δ_{Wiki}=−0.013, 2 seeds). (b) Normalized PPL gap relative to uniform at shared τ values.

multi-epoch exposure: aggressive thresholds retain less data and repeat it more often. Epoch-matched comparisons (rightmost column of Table 8) remove the multi-epoch confound: the U-shape vanishes and PPL decreases monotonically, saturating near the adaptive b^* level at $q \geq 0.5$. Figure 5 visualizes the two curves.

Method	ρ=0.4	ρ=0.6
Uniform	17.694	17.766
Adaptive b^*	17.689	17.740
Hybrid (q>0.95 + adaptive)	17.514	17.530*
Detect & Remove (q>0.5)	17.285	–
Detect & Remove (q>0.3)	17.250	–

Table 7: Method comparison across contamination levels (seed 42). *Hybrid at ρ=0.6 filters high-confidence synthetic (q>0.95), reducing effective ρ to ≈0.4; PPL is not directly comparable to ρ=0.6 baselines.

I Downstream Task Evaluation

Table 9 reports downstream benchmark accuracy for adaptive vs. uniform training at ρ=0.4 (seed 42, Qwen3-0.6B). Across multiple seeds, downstream differences are mixed in direction and uniformly within standard error, confirming that adaptive reweighting has no meaningful downstream impact.

J Higher Contamination Details

Table 10 reports results at ρ=0.6. The full five-method ablation preserves the hierarchy from ρ=0.4 across all three seeds: Grid < Adaptive < Uniform ≈ ESS-only < Random.

The ESS-only/Uniform gap collapses from 0.043 at ρ=0.4 to 0.000 at ρ=0.6 (three-seed means:

Thresh.	Ret. %	Epochs	Raw PPL ↓	Ep.-matched ↓
$q > 0.1$	15.0%	∼6.6	17.40	18.04
$q > 0.2$	∼23%	∼4.3	17.27	–
$q > 0.3$	31.5%	∼3.2	**17.250**†	17.74
$q > 0.4$	∼38%	∼2.6	17.263	–
$q > 0.5$	41.4%	∼2.4	17.28	17.65
$q > 0.6$	∼46%	∼2.2	17.30	–
$q > 0.7$	49.8%	∼2.0	17.32	17.62
$q > 0.8$	∼56%	∼1.8	17.34	–
$q > 0.9$	61.6%	∼1.6	17.42	17.628

Table 8: D&R threshold sensitivity (ρ=0.4, seed 42 unless noted). Raw PPL traces a U-shaped curve (minimum at q>0.3); epoch-matched PPL (single content epoch) decreases monotonically and saturates near the adaptive b^* level (≈17.65) at $q \geq 0.5$, confirming the U-shape is a multi-epoch artifact. † 3-seed mean 17.250 ± 0.001.

Benchmark	Metric	Adaptive	Uniform	Δ
Seed 42 (checkpoint resume)				
ARC-Challenge	acc_norm	0.343	0.348	−0.005
ARC-Easy	acc	0.650	0.657	−0.007
HellaSwag	acc_norm	0.479	0.484	−0.005
PIQA	acc	0.687	0.686	+0.001
WinoGrande	acc	0.578	0.586	−0.008
From-scratch 2-seed mean (seeds 123, 456)				
Mean Δ across 5 tasks				+0.001
Direction				mixed (seed 123: 3/5−; seed 456: 3/5+)

Table 9: Downstream benchmarks (Qwen3-0.6B, ρ=0.4). Seed 42 used a checkpoint-resumed adaptive model; from-scratch seeds (123, 456) show mixed direction with mean $|\Delta| \leq 0.01$. All differences within standard error (±0.01). Models use τ=0.8; τ=0.7 yields identical 3-seed means (0.5414).

both 17.763); ESS binding rises to ∼49% of batches (from ∼43% at ρ=0.4), and the constraints become nearly balanced rather than α_{eff}-dominant. The adaptive method maintains its advantage (Δ=−0.032, consistent across seeds) through the α_{eff} quality constraint, which operates independently of the ESS ceiling. Random reweighting (17.861±0.025) is substantially worse than Uniform, with the highest cross-seed variance, confirming that uninformed weights add noise without corrective signal. The Adaptive-vs-Grid gap is 0.054 at ρ=0.4 (mean-to-mean, consistent with Table 2) and 0.097 at ρ=0.6 (adaptive 3-seed mean vs. grid seed 42; the ρ=0.6 grid result is preliminary and may not generalize across seeds).

K α_{eff} Proxy Validation

Table 11 reports the correlation between $\alpha_{\text{eff}}(0)$ (the detector-based proxy at b=0) and the true batch-level real fraction α_{true}, computed over all training batches at ρ=0.4 (seed 42).

The α_{eff} distribution at B=4 is concentrated:

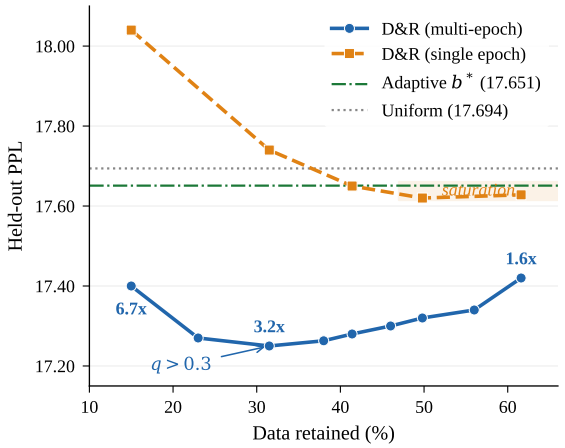


Figure 5: D&R perplexity as a function of data retained. **Blue**: raw (multi-epoch) PPL traces a U-shaped curve; annotations show effective epoch count. **Orange**: epoch-matched (single content epoch) PPL decreases monotonically and saturates at $q \geq 0.5$, matching the adaptive b^* level (green). The raw U-shape is entirely a multi-epoch artifact: controlling for data exposure eliminates it.

Method	Seed 42	Seed 123	Seed 456	Mean
Grid (b=0.3)‡	**17.634**	–	–	–
Adaptive b^*	17.735	17.727	17.732	17.731 ± .004
Uniform	17.766	17.754	17.770	17.763 ± .008
ESS-only	17.770	17.760	17.760	17.763 ± .006
Random $\mathcal{U}(0, 2)$	17.832	17.870	17.880	17.861 ± .025

Table 10: Results at ρ=0.6 (Qwen3-0.6B, $n_{\min}$=2). Adaptive uses τ=0.7 (consistent with main results); other methods are τ-independent. ‡Post-hoc oracle; seed 42 only. Adaptive vs. Uniform: Δ=−0.032, paired t=−15.4, p=0.004 (full precision; 3dp rounding yields t=−9.95, p=0.010). ESS-only collapses onto Uniform (both 17.763); Random (17.861) confirms detector signal is load-bearing.

α_{eff}=0 (11%), 0.25 (34%), 0.5 (36%), 0.75 (17%), 1.0 (3%). MAE decreases with batch size (0.114 → 0.075) as expected from averaging, but the rank-order information that drives b^* selection is already well-captured at B=4.

Cross-seed stability. Table 12 reports the correlation between α_{eff} and the batch mean q-score ($\bar{q}$, computed from known labels) across all three seeds. Since $\bar{q}$ is computed directly from batch composition, this correlation measures the proxy's fidelity without thresholding noise.

L Detector Comparison

Table 13 compares three detector families for adaptive reweighting. The discriminative TF-IDF+LR

B	Pearson r	Spearman r	MAE	Bias
4	0.706	0.699	0.114	-0.029
8	0.697	0.683	0.097	-0.027
16	0.707	0.691	0.075	-0.031

Table 11: Correlation between $\alpha_{\text{eff}}(0)$ and true batch-level real fraction across batch sizes (ρ=0.4, seed 42). Pearson $r \approx 0.70$ is stable across batch sizes. The consistent negative bias (≈ -0.03) indicates that α_{eff} slightly underestimates the true real fraction, biasing the method toward conservative (less aggressive) intervention. At B=4, α_{eff} takes only five discrete values ($\{0, 0.25, 0.5, 0.75, 1\}$), yet rank correlation is preserved.

Seed	Pearson $r(\alpha_{\text{eff}}, \bar{q})$	Spearman ρ	b^*=0 (%)
42	0.8019	0.8761	5.8
123	0.7994	0.8692	5.7
456	0.8006	0.8749	5.9
Spread	0.0025	0.0069	0.2

Table 12: Per-seed α_{eff}–$\bar{q}$ correlation (ρ=0.4, B=4). Cross-seed spread is <0.003 for Pearson r and <0.007 for Spearman ρ, confirming that the proxy's fidelity is stable across random seeds. $r(\alpha_{\text{eff}}, \bar{q}) = 0.80$ is an exact correlation with the mean detector score; the ground-truth correlation $r(\alpha_{\text{eff}}, \text{true real fraction}) = 0.706$ (Table 11) is lower due to thresholding noise in r_i. The per-seed table demonstrates cross-seed stability rather than superseding the ground-truth estimate.

detector (ECE = 0.048) produces the only improvement over uniform. The TF-IDF vs. GPT-2 contrast (comparable AUC: 0.94 vs. 0.91, but 2.7× ECE gap: 0.048 vs. 0.131) isolates calibration as one necessary condition; however, N-gram's better ECE (0.069) does not compensate for its low AUC (0.73), confirming that both sufficient discrimination and good calibration are required. TF-IDF+LR's q-scores have reliable probabilistic semantics, so the α_{eff} proxy accurately tracks batch quality. Generative detectors assign high q-scores to fluent, predictable human text (false positives concentrated among high-quality samples), systematically inflating α_{eff} for clean batches and deflating it for contaminated ones, which inverts the minimum-intervention logic.

M Detector and Synthetic Data Details

TF-IDF+LR detector. TF-IDF features (unigrams + bigrams, max 50,000 features, sublinear TF) fed to Logistic Regression (C = 1.0, L2 penalty, solver = lbfgs, max_iter = 100). Trained on a held-out labeled split (10K documents per class).

Detector	**Type**	**AUC**	**ECE**	**Mean PPL**	**Δ**
TF-IDF+LR	Discriminative	0.94	0.048	$17.650_{\pm.010}$	-0.044
GPT-2 PPL	Generative	0.91	0.131	$17.792_{\pm.005}$	$+0.098$
N-gram PPL	Generative	0.73	0.069	$17.823_{\pm.003}$	$+0.129$

Table 13: Detector comparison for adaptive reweighting (ρ=0.4, τ=0.8, 3 seeds). Uniform baseline PPL = 17.694. Δ = Adaptive − Uniform (negative is better). TF-IDF+LR calibrated via temperature scaling. The τ confound is conservative: TF-IDF+LR at τ=0.7 achieves 17.674 (Table 2).

Post-hoc temperature scaling applied when uncalibrated ECE exceeds 0.05: a scalar T is searched via `scipy.optimize.minimize_scalar` over $[0.1, 10.0]$ on a 20% stratified validation split, yielding T=0.204 (ECE reduced from 0.12 to 0.05). The detector is selected automatically from six configurations as the first satisfying AUC $\in [0.85, 0.98]$ and mid-range score ratio $\geq 5\%$.

Synthetic data generation. Continuations generated with Qwen3-0.6B (self-generation) using nucleus sampling (p=0.95, T=0.8), producing up to 384 tokens per continuation from a 128-token prefix prompt (512 total), targeting ∼100M synthetic tokens. Source prompts are the first 128 tokens of randomly sampled FineWeb documents, used as prefix continuations. Injected at token-level ratio ρ via uniform random mixing across the training corpus.

N Low Contamination Analysis

At ρ=0.2, the adaptive method degrades by +0.046 PPL relative to uniform (0.26% relative, seed 42). The detector's mean q-score at this contamination level is 0.393 (vs. 0.487 at ρ=0.4), indicating that many clean samples receive moderate q-scores, blurring the provenance signal. The regime decomposition reveals 13.7% clean batches (b^*=0), 60.1% α_{eff}-binding, and 26.3% ESS-binding, with mean b^*=1.735 in reweighted batches. The high α_{eff}-binding rate (60.1% vs. 50.9% at ρ=0.4) means the method actively intervenes rather than defaulting to b^*=0, but the unreliable signal causes it to suppress useful training data. This is consistent with the generative detector failures: when q-scores do not reliably separate provenance, the minimum intervention framework amplifies the error rather than correcting it. The effective range of the method is $\rho \geq 0.4$ with $\tau \in [0.7, 0.8]$ (§5.3); lower τ values may extend the range but have not been tested at ρ=0.2.

O Cross-Model Transfer

Preliminary single-seed results for Llama-3.2-3B as a cross-generator (synthetic data generated by Llama, training on Qwen3-0.6B) are shown in Table 14. Results are mixed: MMLU improves while HellaSwag slightly degrades, consistent with the pattern observed in Qwen3-1.7B downstream evaluation (§5.2). Perplexity comparison omitted due to inconsistent evaluation configurations; checkpoints were not retained for re-evaluation.

Metric	Uniform	Adaptive	Δ
MMLU (5-shot)	51.85% ± .41	53.42% ± .41	+1.57pp
HellaSwag (10-shot, acc_norm)	73.08% ± .44	72.49% ± .45	−0.60pp

Table 14: Cross-model transfer: Llama-3.2-3B generator, Qwen3-0.6B trainer (ρ=0.4, single seed). Mixed results reflect evaluation-set distribution sensitivity rather than consistent quality differences.

P Checkpoint Resume Artifact

The checkpoint-resume artifact described here affected early τ=0.8 experiments only. All results in Tables 2–1 use independent from-scratch training (τ=0.7, three seeds).

The adaptive model for seed 42 in early τ=0.8 runs was trained with a checkpoint-resumed run whose first ∼1,000 steps used different hyperparameters (learning rate warmup, gradient accumulation configuration) than seeds 123 and 456, which were trained from scratch. The resume artifact inflates the in-distribution effect (Δ=−0.041 resume vs. Δ=−0.005 from-scratch at τ=0.8) and reverses the out-of-distribution direction:

Domain	Resume	From-scratch	Dir.
FineWeb (in-dist)	−0.041	−0.005	same
OpenWebText (OOD)	+0.058	−0.005	reversed
Wikipedia (OOD)	+0.137	−0.009	reversed

Δ relative to from-scratch uniform (Table 6). Resume: seed 42 original checkpoint. From-scratch: clean replication (D110).

The resume model showed apparent distribution matching (OOD degradation), which was the basis for the "distribution matching" finding in earlier analyses. From-scratch retraining eliminates this pattern: all three evaluation domains show small, directionally consistent improvements, providing no evidence of distribution matching. This underscores the importance of controlling for training artifacts when interpreting small effect sizes across evaluation domains.

Q Additional Contamination Rate

To probe the boundary between the degradation regime (ρ=0.2, Δ=+0.046) and the improvement regime (ρ=0.4, Δ=−0.020), we run adaptive b^* and uniform baselines at ρ=0.3 using the same experimental setup as the main experiments (Qwen3-0.6B, 132M FineWeb tokens, τ=0.7, $n_{\min}$=2).

Method	Seed	FineWeb ↓	OWT ↓	Wiki ↓
Uniform	42	17.422	18.113	15.204
Adaptive b^*	42	17.422	18.123	15.172
Uniform	123	17.433	18.126	15.204
Adaptive b^*	123	17.435	18.148	15.165
Uniform	456	17.419	–	–
Adaptive b^*	456	17.424	–	–
Δ (3-seed mean)	–	+0.002	–	–

Table 15: Results at ρ=0.3 (Qwen3-0.6B, 3 seeds). Holdout Δ=+0.002 ($t(2)$=1.60, p=0.25, non-significant); OOD available for seeds 42 and 123 only, mixed in direction.

The holdout difference is effectively zero (Δ=+0.002, $t(2)$=1.60, p=0.25, non-significant), placing the transition boundary between the degradation regime (ρ=0.2, Δ=+0.046) and the improvement regime (ρ=0.4, Δ=−0.020) within $[0.3, 0.4]$ (Figure 6).

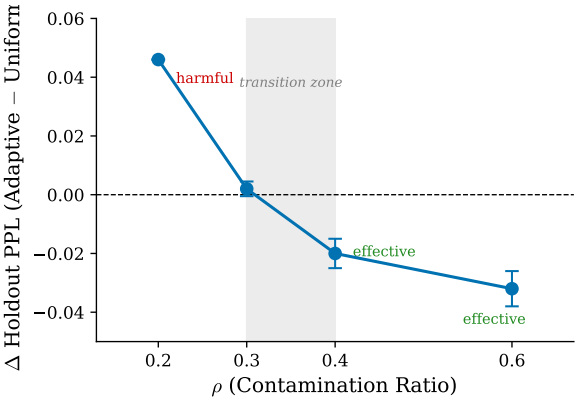


Figure 6: Effect of contamination ratio on adaptive reweighting effectiveness. Δ = adaptive − uniform holdout PPL (lower is better). Error bars show ±1 SD of paired differences across seeds. The ρ=0.2 point (open circle) is from a single seed; all others are 3-seed means. The method transitions from harmful to beneficial between ρ=0.3 and 0.4 (shaded transition zone).

At ρ=0.3, 28% of micro-batches have b^*=0 (clean, no intervention), compared to ∼2% at ρ=0.4; the low intervention frequency explains why the method neither helps nor harms. The

progression across contamination levels—ρ=0.2 harmful (+0.046), ρ=0.3 neutral (+0.002), ρ=0.4 beneficial (-0.020)—shows a graceful transition rather than abrupt failure, consistent with the detector's signal-to-noise ratio gradually crossing the threshold needed for reliable α_{eff} estimation.